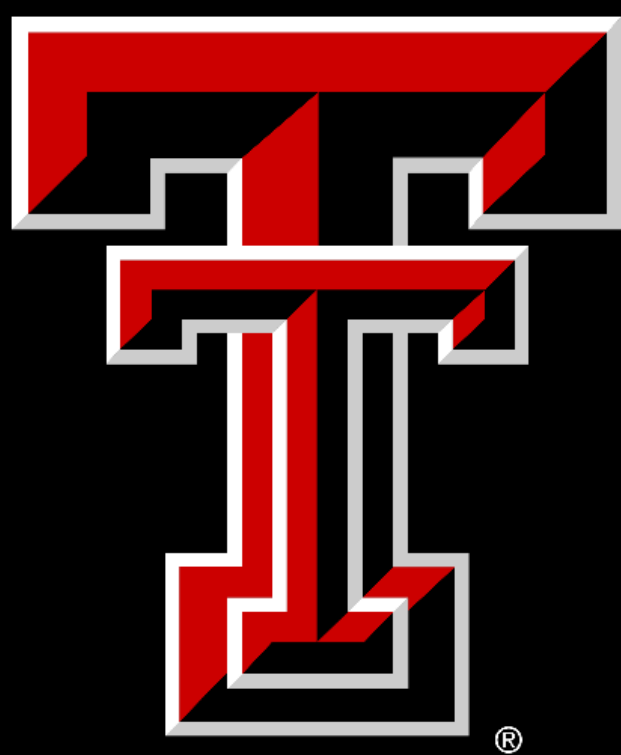


PlastiVis: An Interactive Visualization Tool for Synaptic Networks

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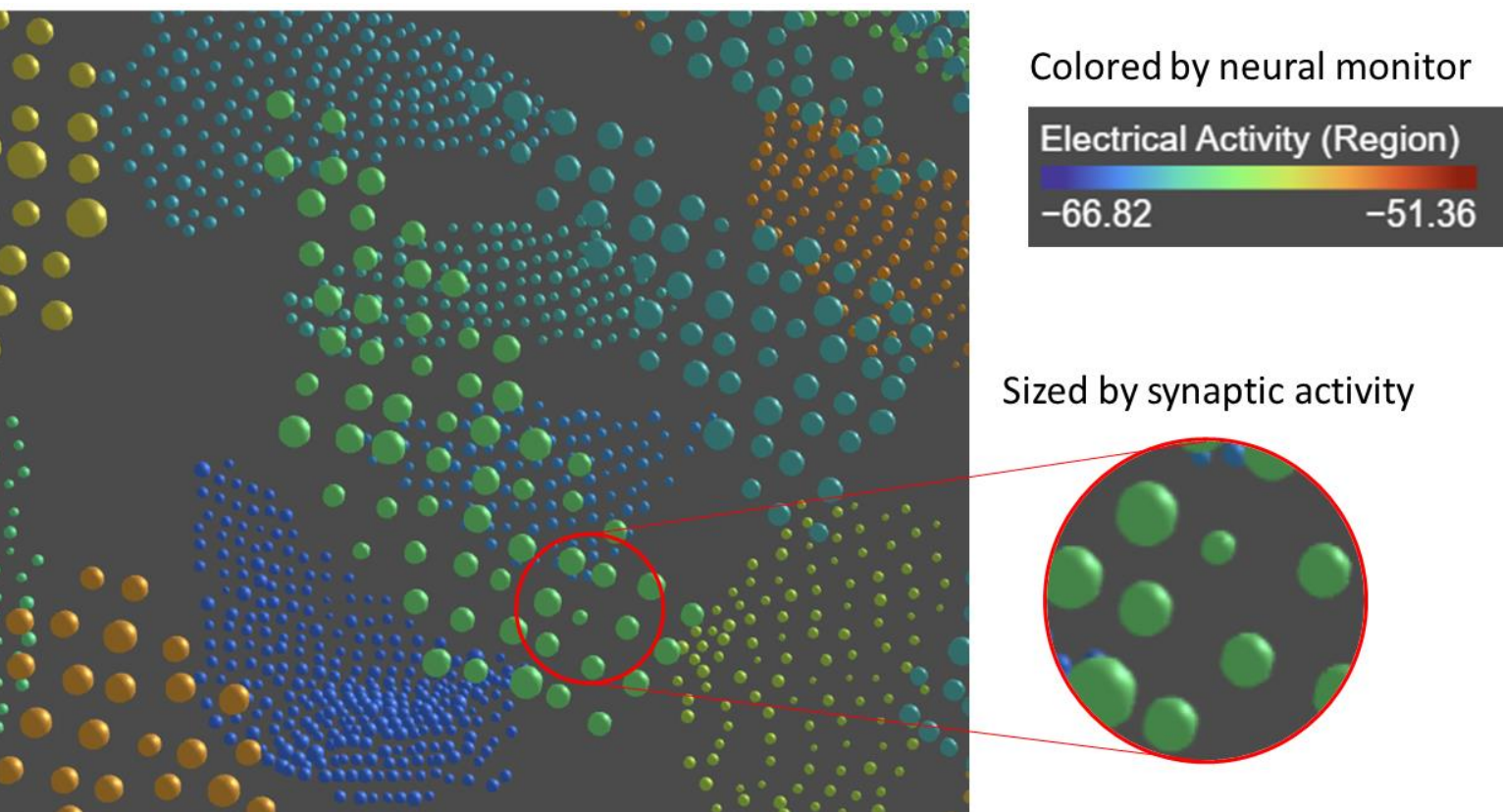
Background

Analyzing neural connections – why/how they are formed and their behavior in different circumstances – is crucial to develop an understanding of core brain processes.

We developed an interactive neural plasticity visualization platform tailored towards dynamic synaptic networks, available on an easily accessible web interface. PlastiVis provides an overview of the data with a 3D visualization alongside a 2D panel to assess change over time for a more detailed analysis of synaptic activity

Data Preprocessing

- Data for four neuronal network simulations provided
- Covered 50,000 neurons with 1,000,000 timesteps
- Reprocessed neuronal data into clusters based on brain area
- Filtered synapses to only highlight creation and deletion events

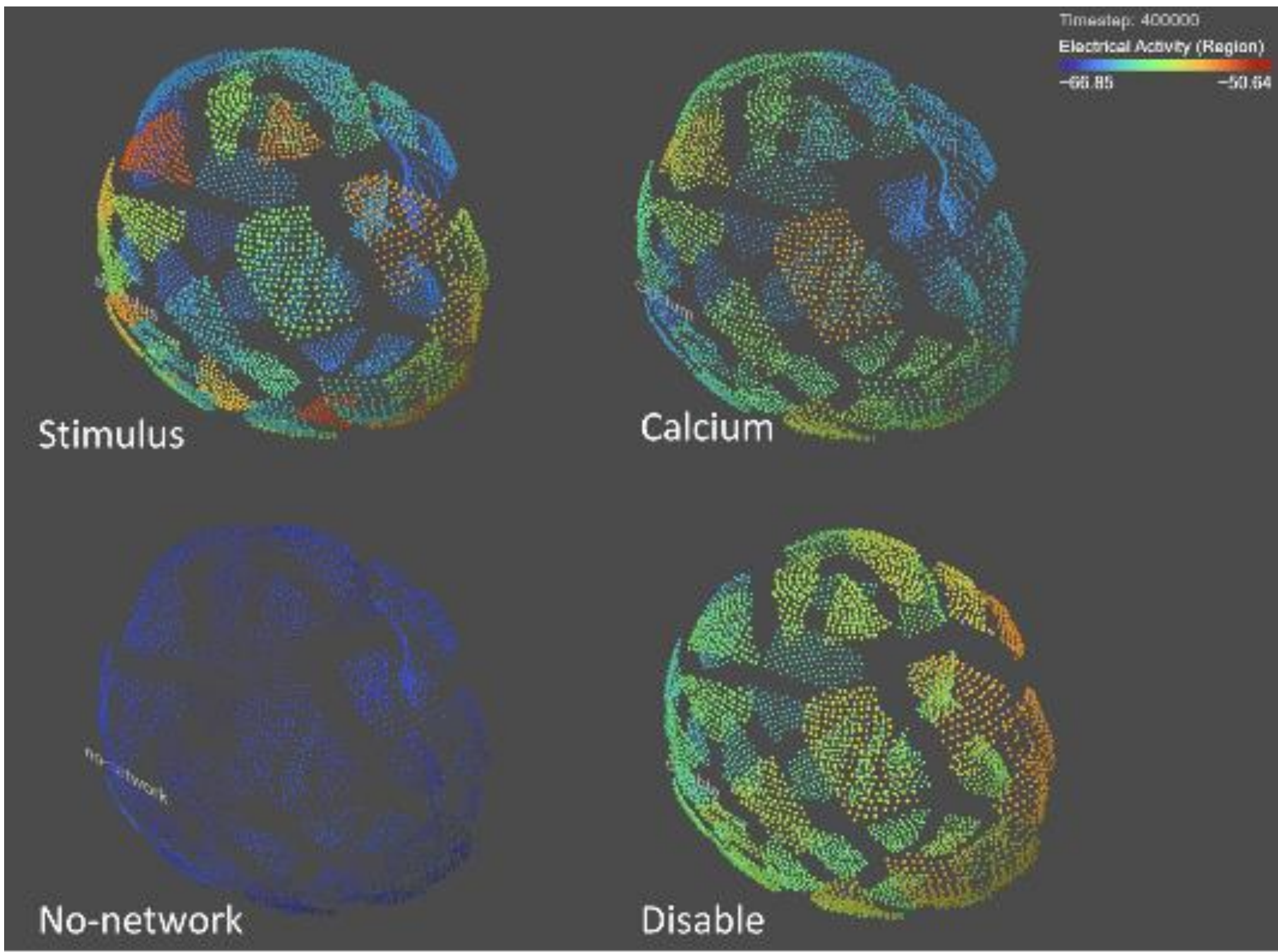
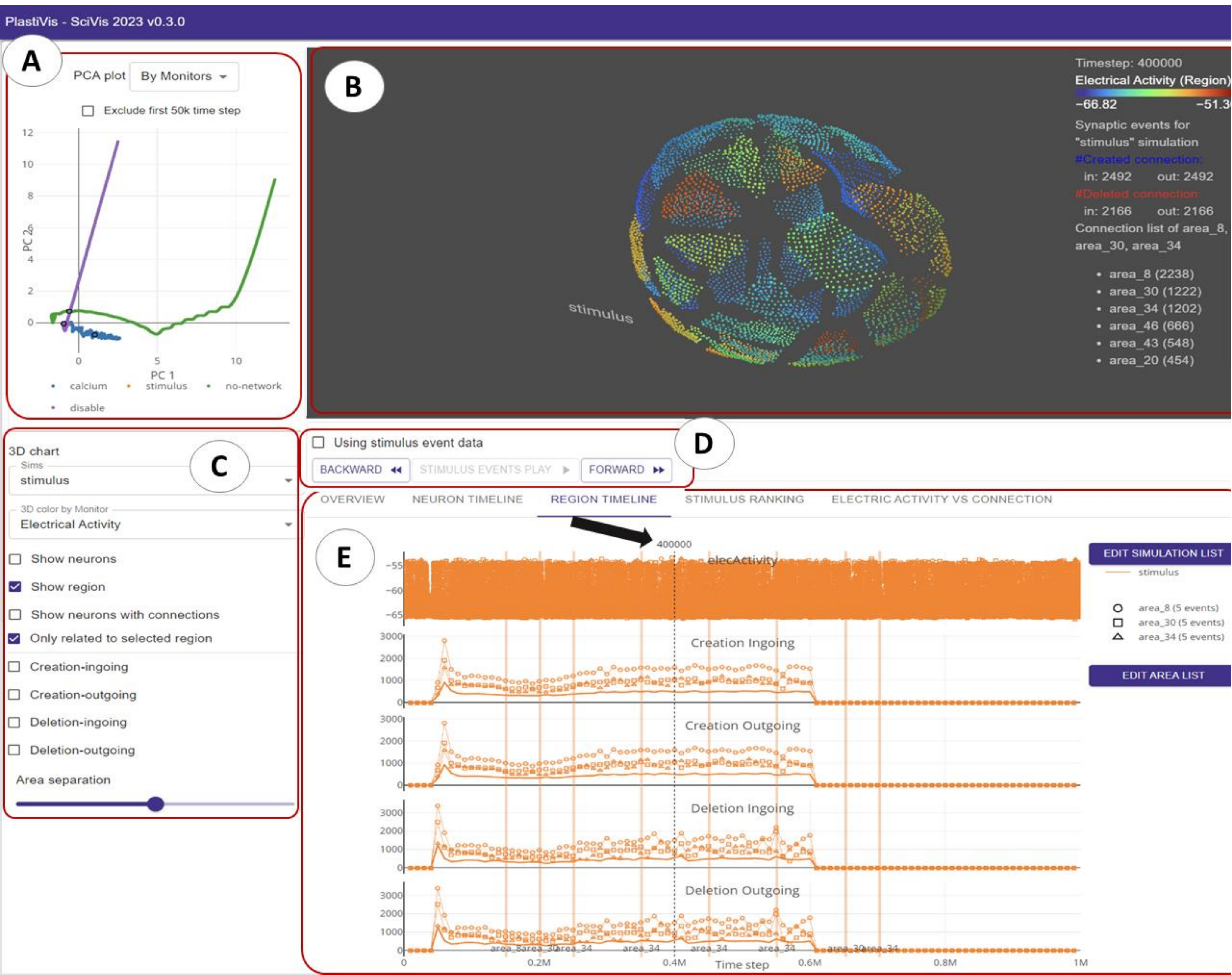


Visual Embedding

- Nodes colored based on selected variable and timestep
- Larger nodes are more synaptically active

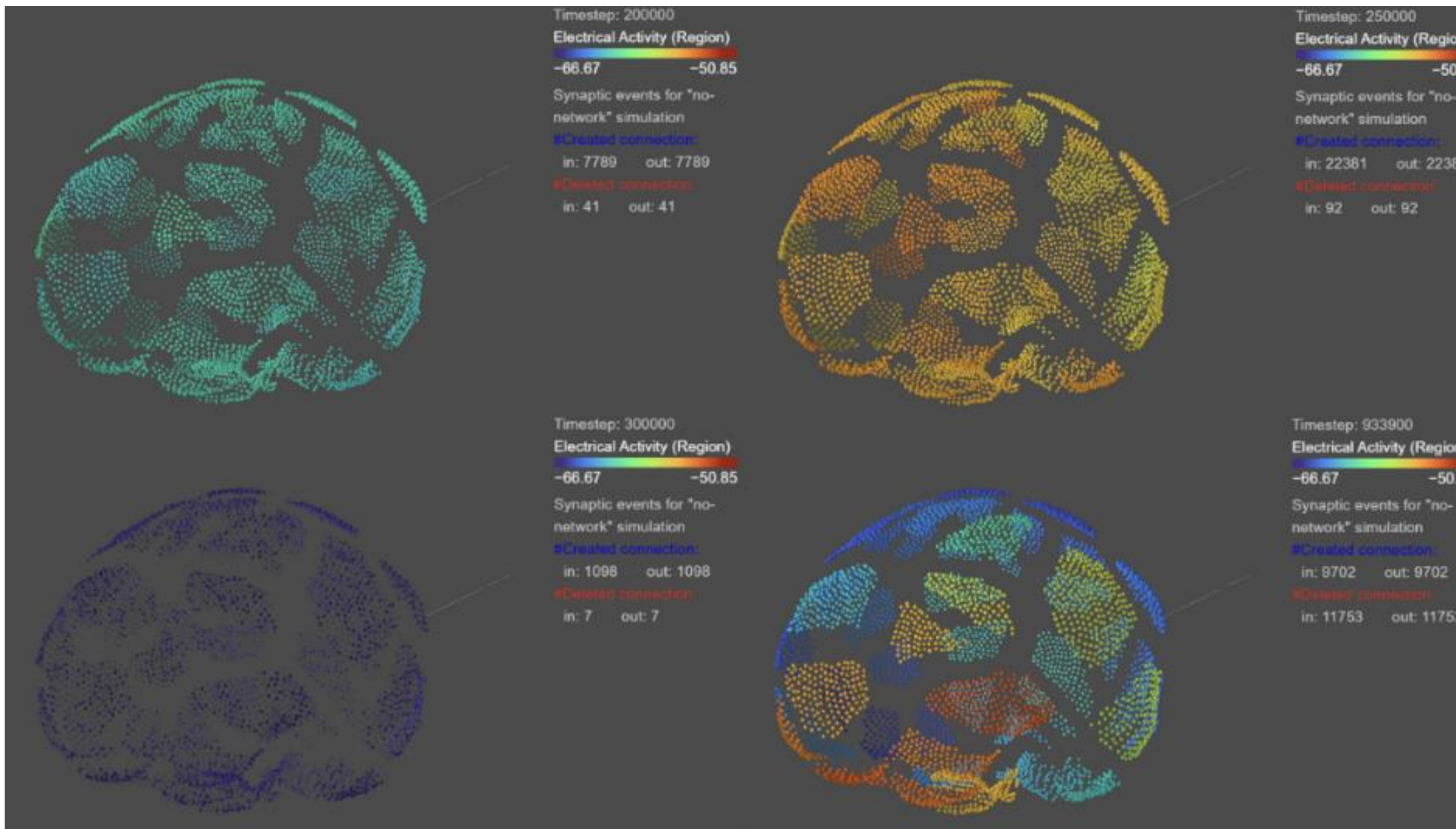
Visualization Design

The overview of the PlastiVis interface with the (A) PCA plot, (B) 3D visualization window, (C) user controls for the 3D chart, (D) dynamic video viewer, and (E) the 2D figure panel hosting a variety of time series and analysis plots.



3D Brain Viewer

- PlastiVis can display one dataset, or compare multiple at the same time (as shown above)
- For a single dataset, the 3D window can show how the variability of neural monitors change overtime spatially
- Summary statistics of synaptic data are displayed under the legend



Temporal Visualization

- PlastiVis supports regional analysis that compares different datasets, regions between different datasets, and regions within a single dataset.
- Each simulation is colored distinctly, and each region is given a unique geometric marker.
- Events of interest (such as stimulation or deletion events are highlighted)
- The time series is synchronized with the 3D visualization, which displays the brain's status at the selected timestep (indicated by the vertical dashed line)
- Selecting regions or time points on the time series graph updates the 3D visualization

Connectivity

We only display newly created and deleted synapses at each time step and connections can be filtered to only regions of interest.

